

CUSTOMER BEHAVIOR ANALYSIS

Alfido Tech Internship Project

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Internship Domain : Data Analysis

Organization : Alfido Tech

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Objective

The objective of this project is to analyze customer behavior using transaction data. This project helps identify customer purchasing patterns, spending behavior, customer segmentation, and churn risk. The analysis also provides business insights and recommendations to improve customer retention and sales performance.

Dataset Description

The dataset used in this project contains customer transaction and behavioral data collected from an e-commerce platform. The dataset includes information such as customer ID, purchase date, product category, product price, quantity, payment method, customer age, gender, returns, and churn status.

Data Cleaning

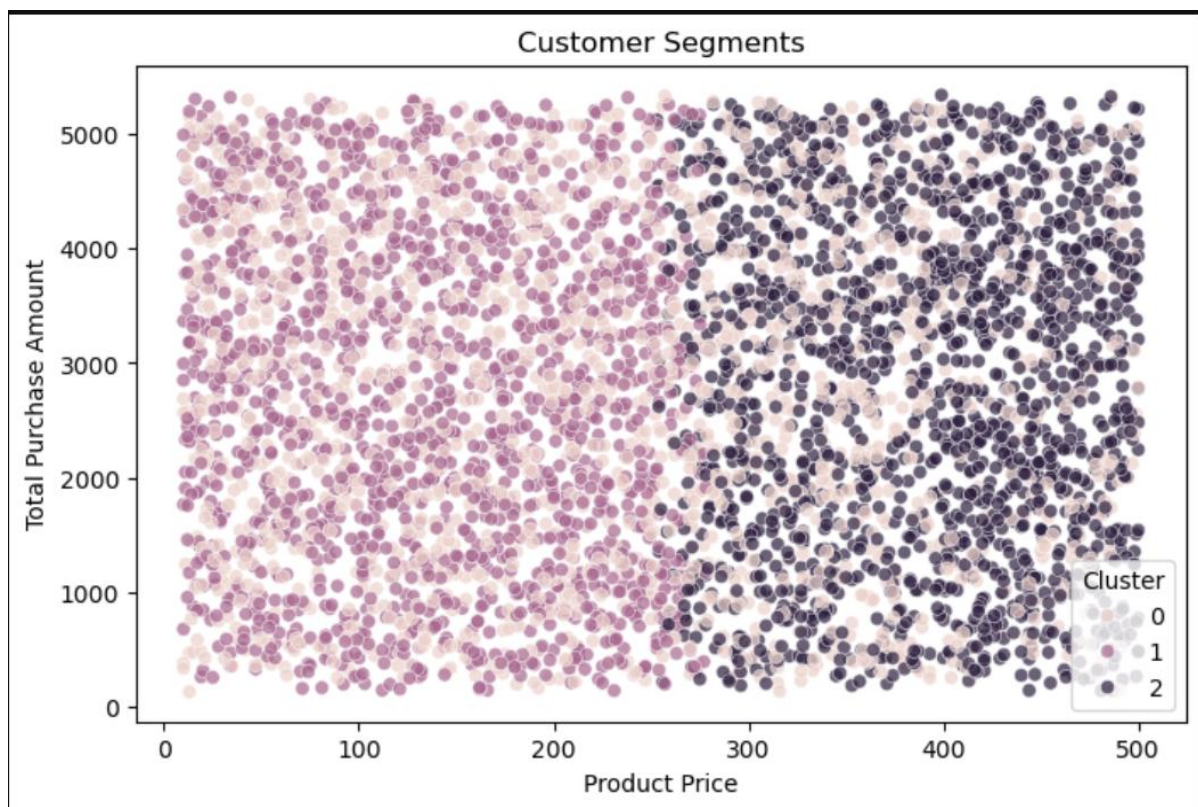
The dataset was checked for missing values, duplicate records, and data consistency issues. Missing values in the Returns column were handled using appropriate data filling techniques. Duplicate records were also verified to ensure data quality and reliability for analysis.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand customer behavior and purchasing patterns. Various visualizations were created to analyze customer age distribution, gender distribution, product categories, payment methods, purchase amounts, and monthly purchase trends.

Customer Segmentation

Customer segmentation was performed using the K-Means clustering algorithm. Customers were grouped into different clusters based on product price, quantity, and total purchase amount. This segmentation helps identify high-value, medium-value, and low-value customers for better business decision-making.

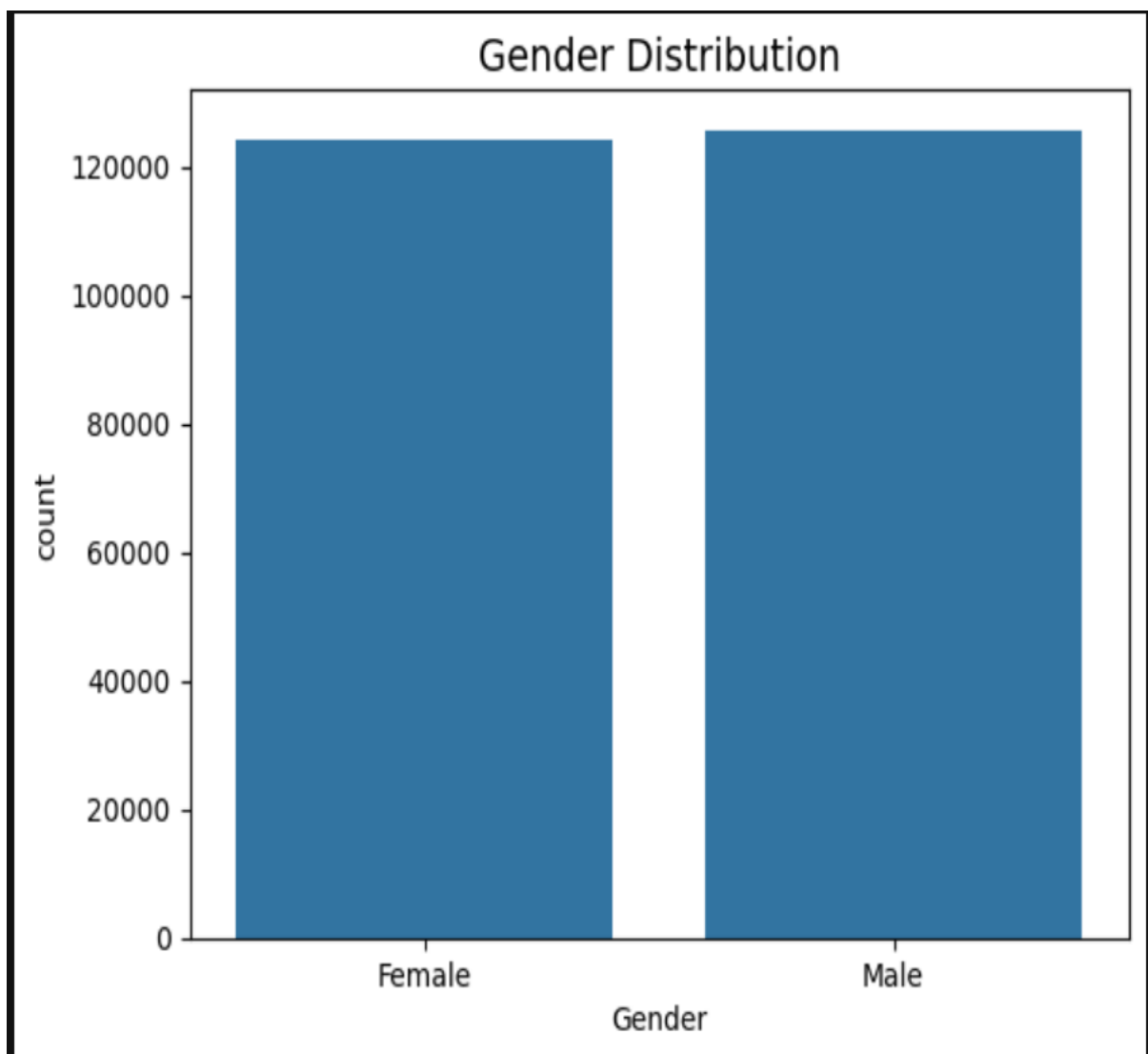


Key Insights

1. Customer purchasing behavior varies across different customer groups.
2. Product categories show balanced customer interest.
3. High-value customers contribute significantly to overall sales.
4. Customer segmentation helps identify different spending patterns.
5. Churn analysis helps businesses identify customers at retention risk.

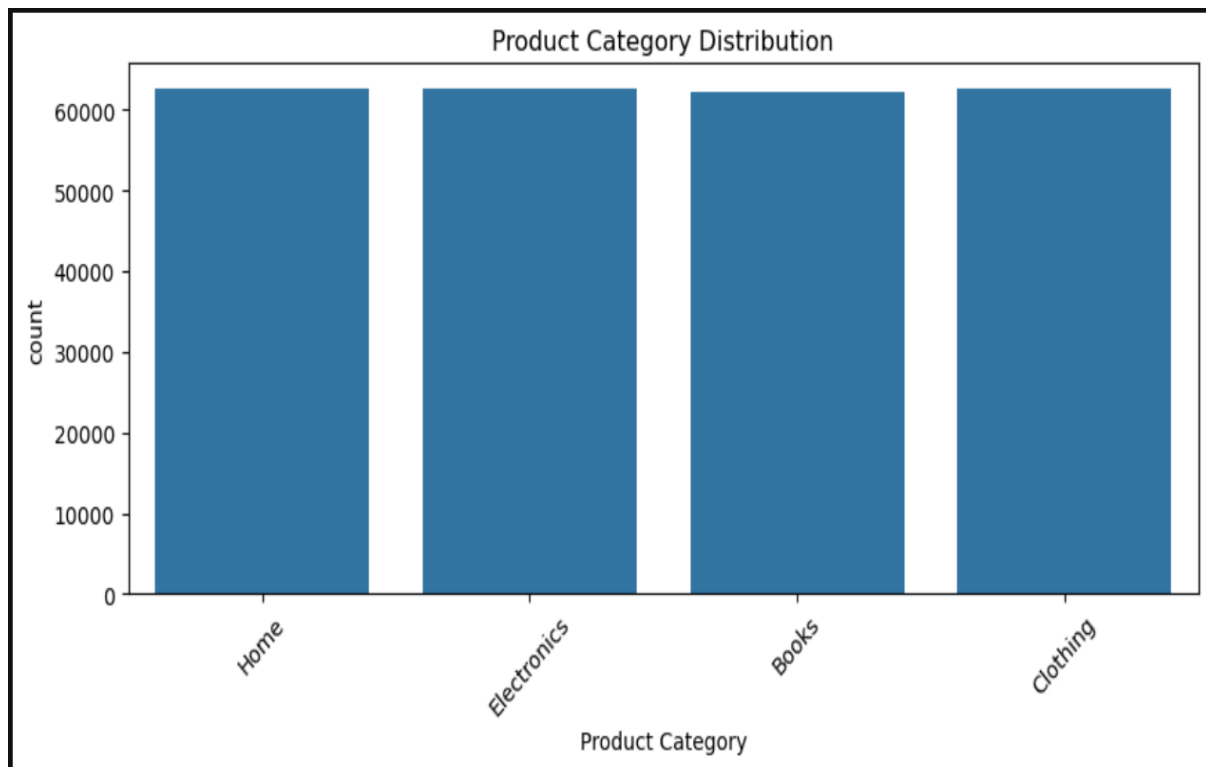
Gender Distribution Analysis

The gender distribution analysis was performed to understand the proportion of male and female customers in the dataset. The visualization below represents the customer distribution based on gender.



Product Category Analysis

Product category analysis was conducted to identify customer interest across different product categories. The graph below shows the distribution of purchases among various product categories.



Conclusion

This project successfully analyzed customer behavior using data analysis and visualization techniques. Customer segmentation and churn analysis provided valuable business insights. The project demonstrates how data analysis can help businesses improve customer retention, understand purchasing behavior, and make better strategic decisions.